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RESEARCH ARTICLE

Hold design supports learning and transfer of climbing fluency

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Abstract

Being a discipline with a broad range of genres, rock climbing is an activity where participants seek to generalize the skills they learn in different performance contexts. A training strategy for achieving skill transfer was explored in a group of experienced climbers. Specifically, we tested the effect of practising on three routes, each of the same difficulty, but where handholds supported opportunities for using either a single technical action or multiple actions. Transfer of climbing fluidity in terms of the geometric index of entropy (GIE) of the hip trajectory was then assessed. We expected that learning would be induced on the route where multiple actions were usable. Results revealed that GIE showed a learning effect only when practice was undertaken on a route designed with multiple graspable edges. Practice on the multi-functional route best explains why the participants' successfully generalized climbing fluency under transfer conditions.

Keywords: *entropy, climbing affordances, skill transfer, motor learning, metastability*

1. Introduction

Rock climbing is a discipline that encompasses a variety of different genres (mountaineering, traditional climbing, sport climbing and bouldering). Due to this diversity many participants seek to transfer (or generalize) their learning between performance contexts. In order to promote participation across genres that manage inherent risks in climbing, there is an important role for developing transferable skills.

Previous research in rock climbing has been shown to facilitate the transfer of climbing fluency to a novel climbing genre (ice climbing) (Seifert, Wattebled, et al., 2013). Experienced rock climbers are able to transfer climbing fluency by reducing behaviours devoted to surface exploration and by using a broader range of movement patterns (Seifert, Wattebled, et al., 2013). Seifert et al. (2013) suggested that a mechanism underpinning the rapid adaptation to a novel environment was in the rock climbers' ability to perceive climbing affordances (invitations for action)

in the ice-fall during their traverse (Seifert, Wattebled, et al., 2013).

Climbing affordances are emergent behavioural opportunities for use of structural features in climbing walls perceived by individuals (Boschker, Bakker, & Michaels, 2002; Pezzulo, Barca, Bocconi, & Borghi, 2010). Climbing affordances are specified by information-based relationships between structural characteristics of a climbing surface relative to structural and functional characteristics of a performer (Seifert et al., 2014). Knowledge of climbing affordances are likely to support: a more rapid adaptation to climbing a new route (Boschker & Bakker, 2002), more fluid climbing (Sibella, Frosio, Schena, & Borghese, 2007), more efficient exploration of the climbing surface (Seifert et al., 2014) and the transfer of climbing fluency (Seifert, Wattebled, et al., 2013). Determining information that specifies climbing affordances requires experience in climbing tasks (Boschker et al., 2002; Seifert, Wattebled, et al., 2013).

Induced movement variability during practice has been widely connected to improved retention and transfer of skills (Chow, 2013; Magill & Hall, 1990; Ranganathan & Newell, 2013; Schöllhorn, Mayer-Kress, Newell, & Michelbrink, 2009). The mechanisms that underpin the ability to reinvest learning to different conditions have been related to increased exploration of different movement patterns and the information-based relationships used to control them (Chow, Davids, Hristovski, Araújo, & Passos, 2011; Huet et al., 2011; Magill & Hall, 1990; Ranganathan & Newell, 2013; Schöllhorn et al., 2009).

In complex systems, metastability is a regime of movement coordination where multiple affordances functionally overlap, specifying varied opportunities for action (Hristovski, Davids, Araújo, & Button, 2006). Metastability emerges when different parts of a movement system become loosely coupled allowing a rapid, functional reorganization between system components to emerge (Kelso, 2012). During practice in a metastable regime, movement patterns are spontaneously explored from trial-to-trial (Hristovski et al., 2006; Hristovski, Davids, Araújo, & Passos, 2011; Pinder, Davids, & Renshaw, 2012). Seifert et al. (2013) detailed an approach for setting up a metastable regime in climbing by using holds designed with edges that allowed a variety of grips to emerge in climbers including: either a slope grip (a grasping technique where the hold is taken with an over-hand grip, similar as one would take the rung of a ladder) or a pinch grip (a technique where the hold is taken with the knuckles running vertically with respect to the ground plane). In the study of Seifert et al. (2013), a group of inexperienced climbers (level 5c French Rating Scale of Difficulty (F-RSD)), practised on a route (level 5c F-RSD) for four trials. It was found that exploration, in terms of touching but not grasping holds, was induced. Exploration resolved with practice and the use of pinch style grasping increased and the ratings of the hold usability also increased (Sanchez & Dauby, 2009; Seifert, Orth, Hérault, & Davids, 2013).

These results suggested that, through inducing exploration of climbing affordances, practice in a metastable regime may support the acquisition of generalizable skills, being evident in learning behaviours and the ability to transfer skill to a different climbing route. In representative learning designs, skill is generalizable from one context to another (such as between a laboratory and a day-to-day setting) on the basis that features in the environment and task, across settings, support similar information-movement relationships (Araújo, Davids, & Passos, 2007).

An unresolved problem in studies of climbing performance behaviours concerns how to understand

skill transfer. Although practice effects have been observed (Boschker & Bakker, 2002; Cordier, Dietrich, & Pailhous, 1996; Cordier, Mendès-France, Bolon, & Pailhous, 1993, 1994; Cordier, Mendès-France, Pailhous, & Bolon, 1994; Seifert, Orth, et al., 2013), studies are yet to have directly considered transfer related to different practice interventions. The aim of this study was to determine whether practice in a metastable regime supports the transfer of climbing fluency to another route.

2. Methods

2.1. Participants

Six participants (mean age = 23.6) were recruited based on their self-reported red-point level under top-rope conditions (level 6a F-RSD). They were observed under informed consent with ethical approval.

2.2. Experimental procedure

Data were collected over 4 days, with at least 2 days separating each session. All sessions involved participants being equipped with a harness and climbing shoes. An light-emitting diode (LED) was fixed to the harness over the sacroiliac joint. Participants then completed a climbing-specific warm-up. They then completed three top-roped climbs. Before each climb, individuals were allowed 3 minutes to visually examine the route (called route preview). Visual inspection during preview involves looking at the route from the ground. Each climb occurred on a different experimental route, the order of which was counterbalanced. Between each climb, a seated 5 minute rest was enforced. On the fourth day of data collection, climbers undertook a transfer test at the end (also with route preview and a top-rope). Data collection was via a fixed video that captured performance during each entire traverse.

2.2.1. Routes. The three experimental routes practised on by all participants, included a horizontal only edged route, a vertical only edged route and a double-edged route (each hold had two graspable edges: a horizontally orientated and a vertically orientated edge). A transfer route was also climbed by all participants and was composed of a mixture of the types of holds used in the experimental routes. Each route was set by an experienced setter and the difficulty level held constant at level 5c. The ratings were confirmed by consensus with two other route setters. All routes were 10.4 m in height and consisted of 20 holds. For a summary of the experimental procedure see Figure 1(A).

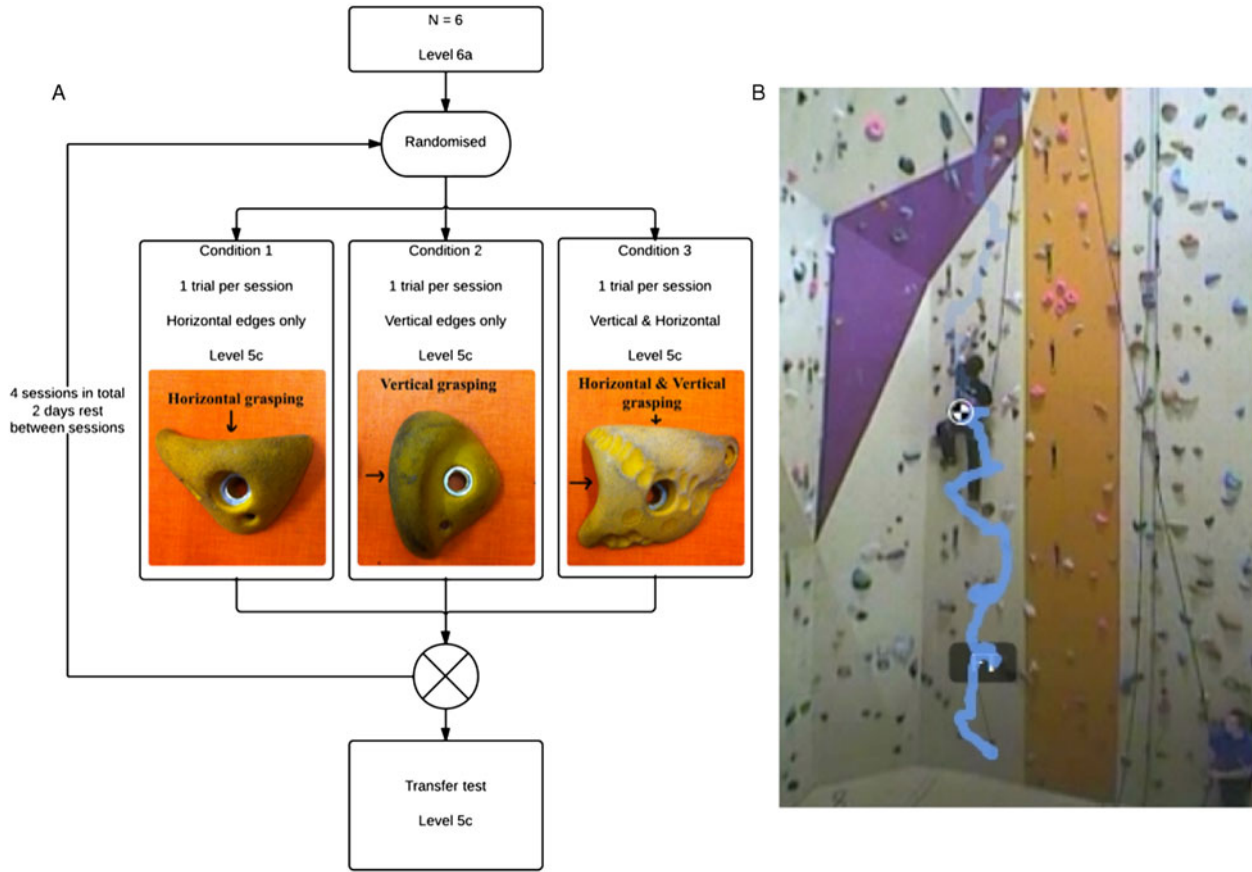


Figure 1. Experimental design and data treatment. (A) The experimental design. (B) The semi-automatic tracking procedure on corrected footage.

2.2.2. Data treatment. The primary variable used to investigate transfer processes after practice was the geometric index of entropy (GIE) of the displacement of the hips in the frontal plane. The GIE is calculated by recording the distance of the path covered by the hips (L) and the perimeter of the convex hull around that path (c).

$$\text{GIE} = \log 2 \left(\frac{2L}{c} \right).$$

According to Cordier, Bolon, et al. (1994), the GIE can assess the amount of fluency of a curve. The higher the entropy, the higher the disorder of the climbing trajectory. Skilful climbing can be reflected in a low entropy value because it is associated with low energy expenditure, less variation in acceleration and use of more technically advanced actions to maintain a more direct path up a vertical surface (Sibella et al., 2007).

To assess the GIE, video footage of each climb was corrected for distortion using Virtual Dub[®] (<http://www.virtualdub-fr.org>). The position of the LED during each climb was then digitized using a semi-automatic tracking procedure with Kinovea[®]

(<http://www.kinovea.org>) (see Figure 1(B)). The horizontal and vertical positions of each entire traverse were then used to determine the GIE for each climb.

Data were statistically treated by using repeated measures ANOVA for the trials (4) and route (3) effects. Planned contrasts were then designed anticipating an interaction effect between route and the rate of change in GIE trial-to-trial. Follow-up t -tests were then used to compare performance data in the final trials on each route to performance on the transfer test.

3. Results

3.1. Main effects

No violation of heterogeneity was detected. There was a significant route effect, $F(2, 10) = 5.49$, $p = .025$, but no significant trial effect, $F(3, 15) = .516$, $p = .68$. A significant route by trial interaction confirmed expectations and suggested that trial effects were present but were dependent on which route the climbers were practising on, $F(6, 30) = 6.43$, $p < .001$.

3.2. Route by trial

The route by trial interaction indicated that trial-to-trial changes in entropy values, associated with practice, differed according to the specific route being practised upon. To break down this interaction, contrasts were performed comparing data from each single edge route (vertical- and double-edged routes) relative to the double-edged route across each trial-to-trial comparison (Trial 1-to-Trial 2, Trial 2-to-Trial 3 and Trial 3-to-Trial 4). The horizontal only to double-edged condition contrasts showed that there was a significant difference between conditions when going from trial 1 to trial 2, $F(1, 5) = 16.195$, $p = .010$. The same trend was revealed when comparing the vertical only to the double-edged condition, with contrasts showing there was a significant difference between conditions when going from trial 1 to trial 2, $F(1, 5) = 40.089$, $p = .001$. Although not part of the planned contrast, in comparing the effect of a different rate of change in entropy between trial 1 and trial 2 between the horizontal only and vertical only conditions, there was no significant difference apparent, $F(1, 5) = .005$, $p = .947$. Overall these data confirmed that practice effects were dependent on the route.

In examining the marginal means of the GIE plotted according to the condition and trials (see Figure 2), it is clear that the rate of change on the double-edged route showed a decreasing trend from

trial 1 to trial 2, as opposed to both the horizontal only and vertical only edge routes that show an immediately lower amount of entropy compared with the double-edged condition, and that remained fairly constant through practice.

3.3. Transfer effect

Dependent t -tests between each condition and the transfer test confirmed that the climbers' entropy values on the horizontal, $t(6) = 0.671$, $p = .532$, vertical, $t(6) = 0.446$, $p = .674$, and double-edged, $t(6) = 0.612$, $p = .567$, routes were not significantly different to the transfer condition.

4. Discussion

The purpose of this study was to determine whether practice under a metastable regime could induce learning and facilitate transfer of climbing fluency on surfaces differing in hold properties.

4.1. The effect of metastable route design on fluency during practice

Data from the single edge only conditions showed that both horizontal grasping and vertical grasping were already stable, acquired behaviours in the sample of climbers that we observed. Furthermore, because the double-edged route induced learning,

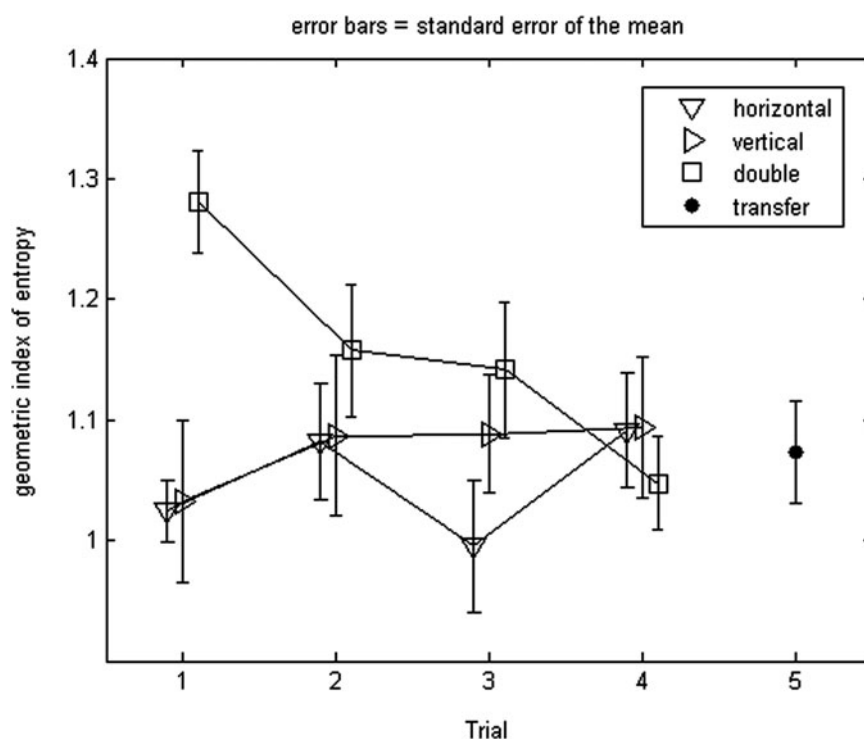


Figure 2. GIE for each trial for each condition.

these findings suggested that enhanced movement fluency may have been due to the availability of choice during route finding. Similar to previous findings by Cordier, Bolon, et al. (1994), participants needed to fine-tune the ways in which they transitioned from one hold to the next through practice.

In the study by Cordier, Bolon, et al. (1994) when individuals were presented with a novel route finding problem, climbers showed an initially elevated GIE, followed by a rapid reduction and asymptote. Their data followed a similar pattern to our data only in the double-edged route condition. The key point of difference between data from our study and those of Cordier, Bolon, et al. (1994), is that our data indicated that not only is route novelty important for inducing learning, but also how route design supports variability in adapting to performance constraints.

4.2. *Transfer of climbing fluency*

On the transfer route, climbing fluency remained at a level no different to the final trial of each of the three routes. These data were interpreted to indicate that it was the design of multiple grasping options in the double edged route that drove the transfer of climbing fluidity. This is because the transfer route was composed of holds with characteristics of each of the three routes. If the climbers had not practised on the double-edged route, climbing fluency would have likely diminished and shown similar levels of entropy as observed on the first practice trial on the double-edged route. This interpretation of the data is supported by the observation that neither the vertical only, nor the horizontal only, routes induced increased entropy at trial 1.

4.3. *Implications and future research*

By making multiple affordances functionally available during practice, learning can be induced without needing to increase the absolute difficulty of a task (the likelihood of failure). This is an important finding in outdoor pursuits, like climbing, where participant safety is a concern (Collins & Collins, 2012; Davids, Brymer, Seifert, & Orth, 2014). When different climbing affordances are made functionally available during route finding, a degree of uncertainty is represented requiring the climber to adapt their performance behaviours. Because learning under these constraints is emergent (not prescribed by instructions), the transfer of skill to a new set of representative task constraints may be facilitated (Davids et al., 2014).

A climbing practitioner can expect that inducing metastability in performance will initially involve

some movement variability that becomes stabilized with practice. The skills acquired that support climbing fluency appear to be available for reinvestment under similar levels of uncertainty during route finding on a new route. From an applied perspective in route design (for example, in commercial, pedagogical, performance contexts), these findings open a new strategy for designing routes that is based on a coherent theoretical framework (Davids et al., 2014).

Further research is needed to consider what sort of capacities for transfer would be observed in individuals who have not already stabilized movement patterns important for fluid climbing when using holds with a horizontal or vertical orientation. Previous work involving beginners has suggested that individuals can rapidly adapt to horizontal climbing edges because they share similar affordances to day-to-day tasks (such as climbing ladders) (Seifert, Orth, et al., 2013; Seifert, Wattebled et al., 2013). Beginners, however, appear to require specific experience in finding affordances in handholds that require different hand and/or body orientations to those of ladder style climbing (Boschker & Bakker, 2002). With respect to beginners, future work can investigate whether metastability in route design is an effective strategy for coaching inexperienced climbers to acquire new skills. It can be anticipated that beginners might benefit from such a strategy because they can use more basic movement patterns as a platform from which to explore affordances that need to be perceived (Davids et al., 2014).

For future research aimed at understanding the mechanisms of transfer because exploration is often implicated in underpinning practice variability effects (Schöllhorn et al., 2009), future research can also consider measurement strategies to quantify exploration. A limitation of GIE is that it is a global indicator of learning (Cordier, Mendès-France, Pailhous, et al., 1994). Additional methods need to be developed to assess behaviours specific to exploration, such as the use of different movement patterns (Seifert, Wattebled, et al., 2013) or the key periods for the use of exploring handholds (Fuss & Niegl, 2008).

Related to the question of observing exploration is whether mechanisms that support transfer due to experience (as opposed to intervention) can also provide insights into more effective training design. Elite climbers (classified with ability levels greater than 7c+ and 8a+ F-RSD, females and males, respectively; see Draper et al. (2011)) are understood to very rapidly adapt to new constraints without displaying 'overt' exploratory behaviours (Fuss & Niegl, 2008; Seifert et al., 2014). As discussed in the Introduction, experienced climbers are able to use information picked-up related to opportunities for

action to more effectively perform in route preview (Boschker et al., 2002; Pezzulo et al., 2010) and route finding tasks (Sanchez, Lambert, Jones, & Llewellyn, 2012). It is possible that an elite group of climbers would immediately adapt to the task presented in this study through exploration that is at different levels of analysis, such as through visual exploration of information during preview.

5. Conclusion

This study showed how informational properties in the indoor climbing context can be used to induce learning and transfer during route finding. Although experienced climbers may have a stable repertoire of movement patterns, it is important that they learn how to adapt these functional skills to climbing on new routes and surfaces. By designing metastability in practice tasks, through designing hand holds so that climbing affordances functionally overlap, learning can be induced and transfer of climbing fluency in route finding can be facilitated.

Disclosure statement

No potential conflict of interest was reported by the authors.

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